PORTFOLIO #4

COMPUTER FUNDAMENTALS AND PROGRAMMING (BES 10a)

2nd Semester (2025-2026)

Name: Van Zan Jave C. Lucila Course, Year & Block: BSChE 1B

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##Read Me: In fulfillment for the subject BES
10a, this project is a Python-based Number
Guessing Hangman game that applies Chapter
5 loop and iteration concepts. It features a
nested setup, using an outer definite "for" loop
to cycle through game rounds and an inner
indefinite "while" loop for active, turn-by-turn
guessing. The game logic relies on basic loop
patterns: a counter variable tracks mistakes , a
boolean search flag verifies guesses, and an
"if/elif" structure prints the text-based
hangman drawings. Finally, "break" statements
handle wins or losses , while a "continue"
statement skips over lines starting with #,
showing how simple control flow can build an
interactive terminal application.

##This is the title section
#Hangman
print("==================NUMBER
GUESSING HANGMAN=================")
print("Instructions: Guess one digit at a time.
Type '#' to write a")
print("note, or 'done' to quit.")
# The list of secret numbers for each round
all_codes = [
    ['3', '7', '4'],
    ['9', '1', '5'],
    ['0', '8', '2'],
    ['6', '4', '8']
]

# Track if the player wants to keep playing the
game
play_again = True
```

```python
# Loop through each secret number list, one
round at a time
for secret_code in all_codes:

    # Stop the game completely if the player
chose to quit
    if play_again is False:
        break

    # Reset the guesses and mistakes at the
start of every new round
    guessed_digits = ''
    mistakes = 0
    max_mistakes = 5

    print()  # Prints a blank line for spacing

print('=====================================
==========================')
    print('🆕 Starting a brand new round with a
new hidden number!')

print('=====================================
==========================')

    # Keep running the current round until it
ends
    while True:
```

```python
    # Print the hangman picture depending on how many mistakes were made
    print()  # Prints a blank line for spacing
    if mistakes == 0:
        print("  +---+")
        print("  |   |")
        print("      |")
        print("      |")
        print("      |")
        print(" =======")
    elif mistakes == 1:
        print("  +---+")
        print("  |   |")
        print("  O   |")
        print("      |")
        print("      |")
        print(" =======")
    elif mistakes == 2:
        print("  +---+")
        print("  |   |")
        print("  O   |")
        print("  |   |")
        print("      |")
        print(" =======")
    elif mistakes == 3:
        print("  +---+")
        print("  |   |")
        print("  O   |")
        print(" /|   |")
        print("      |")
        print(" =======")
```

```
        elif mistakes == 4:
            print("   +---+")
            print("   |   |")
            print("   O   |")
            print("  /|\\  |")
            print("       |")
            print("  =======")
        elif mistakes == 5:
            print("   +---+")
            print("   |   |")
            print("   O   |")
            print("  /|\\  |")
            print("  / \\  |")
            print("  =======")

        print('--------------------------------------------------
')
        print('Current Hidden Number:')
        code_cracked = True

        # Go through each digit of the secret
number one by one
        for digit in secret_code:

            # Check if this specific digit was
already guessed
            found_digit = False
            for correct in guessed_digits:
                if correct == digit:
                    found_digit = True
```

```
            # Show the digit if it was guessed,
otherwise show a blank space
            if found_digit is True:
                print(digit)
            else:
                print('_')
                code_cracked = False

        print('--------------------------------------------------
')

        # End the round if the player successfully
guessed all the digits
        if code_cracked is True:
            print('🎉 Great job! You guessed the
full number and won this round!')
            break

        # End the round if the player runs out of
guesses
        if mistakes == max_mistakes:
            print('🔒 Game Over! The hangman is
fully drawn.')
            print('The correct number was:',
secret_code)
            break

        guess = input('Guess a digit: ')
```

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        # Skip everything else and ask again if the line starts with '#'
        if len(guess) > 0 and guess[0] == '#':
            print('💭 Saved your note. Moving to the next turn.')
            continue

        # Stop the whole game if the user types 'done'
        if guess == 'done':
            play_again = False
            break

        # Check if the player's guess matches any digit inside the secret number
        correct_guess = False
        for digit in secret_code:
            if guess == digit:
                correct_guess = True

        # Save correct guesses or add to the mistake count
        if correct_guess is True:
            print('✨ Nice! That digit is in the number.')
            guessed_digits = guessed_digits + guess
        else:
            print('❌ Wrong guess! An item is added to the hangman drawing.')
            # Add 1 to the total mistakes count
            mistakes = mistakes + 1

        print()  # Prints a blank line for spacing
```

```python
print('==================================
==========================')

print()  # Prints a blank line for spacing
print('Game over! Thanks for playing!')
```

CODE DEMO:

WORKING CODE

```
==================NUMBER GUESSING HANGMAN==================
Instructions: Guess one digit at a time. Type '#' to write a
note, or 'done' to quit.

============================================================
🆕 Starting a brand new round with a new hidden number!
============================================================

   +---+
   |   |
       |
       |
       |
  =======
------------------------------------------------------------
Current Hidden Number:
_
_
_
------------------------------------------------------------
Guess a digit: 4
✨ Nice! That digit is in the number.

============================================================

   +---+
   |   |
       |
       |
       |
  =======
------------------------------------------------------------
Current Hidden Number:
_
_
4
------------------------------------------------------------
Guess a digit: 7
✨ Nice! That digit is in the number.

============================================================

   +---+
   |   |
       |
       |
       |
  =======
------------------------------------------------------------
Current Hidden Number:
_
7
4
------------------------------------------------------------
Guess a digit: 8
❌ Wrong guess! An item is added to the hangman drawing.

============================================================

   +---+
   |   |
   O   |
       |
       |
  =======
```